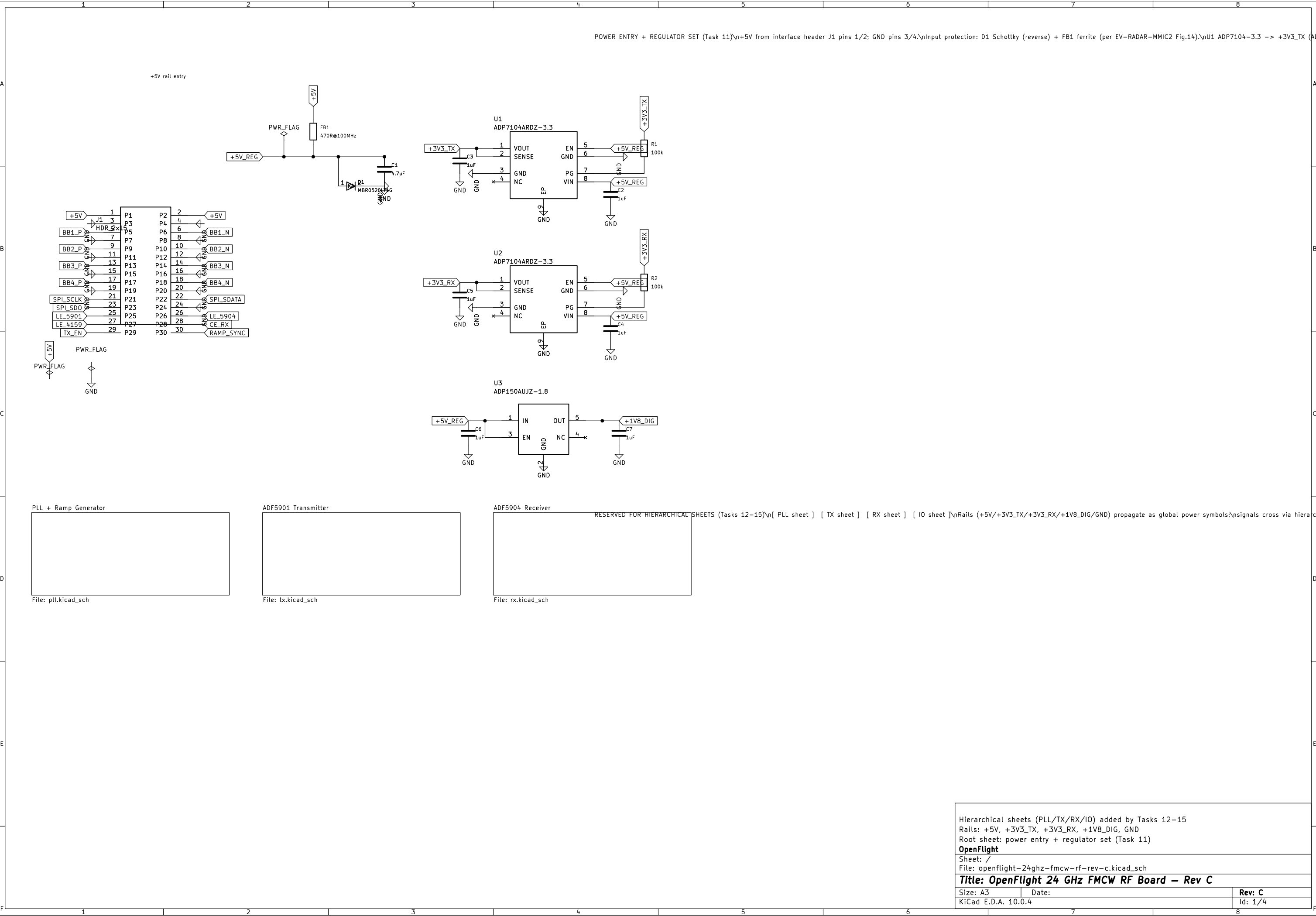
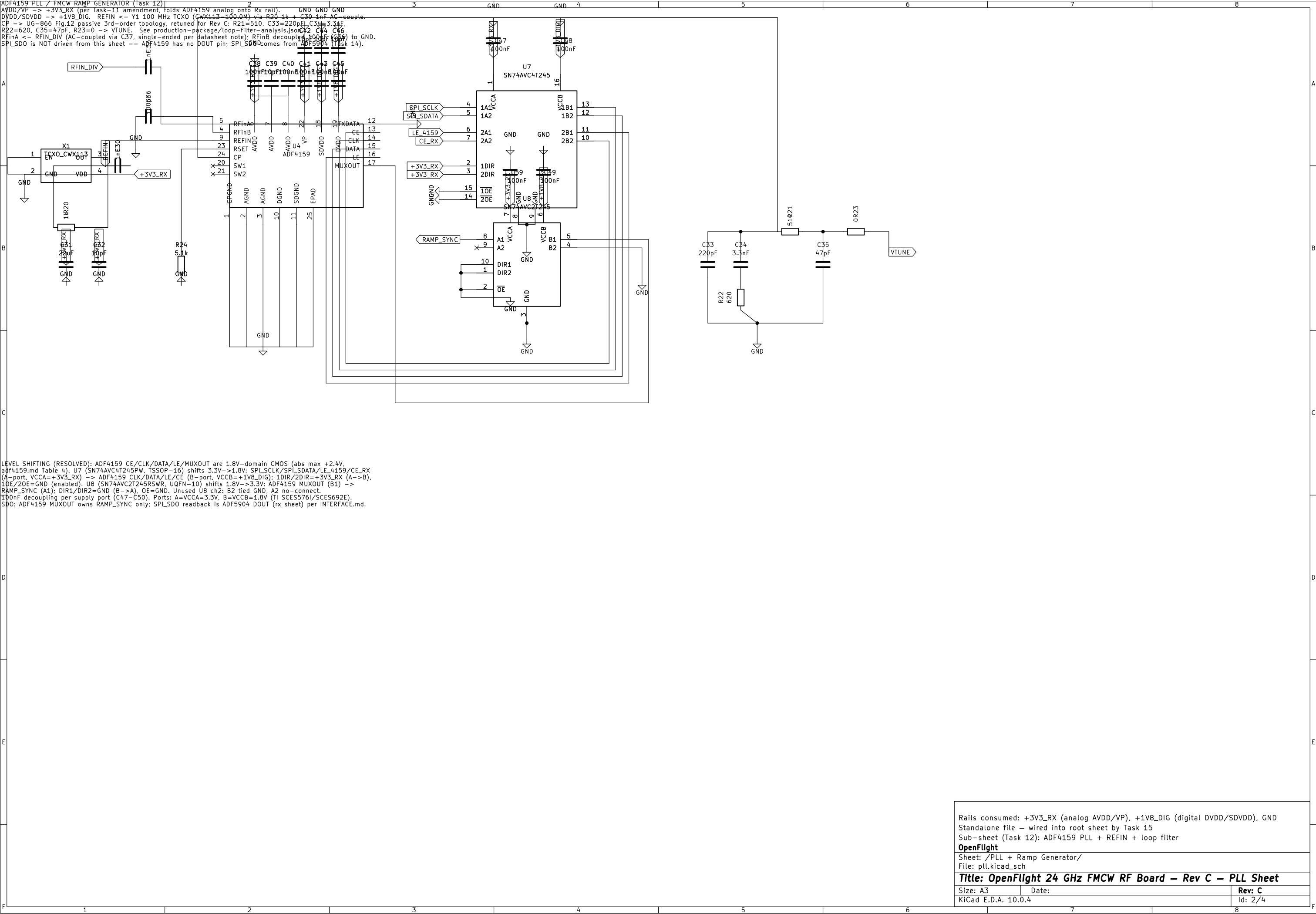


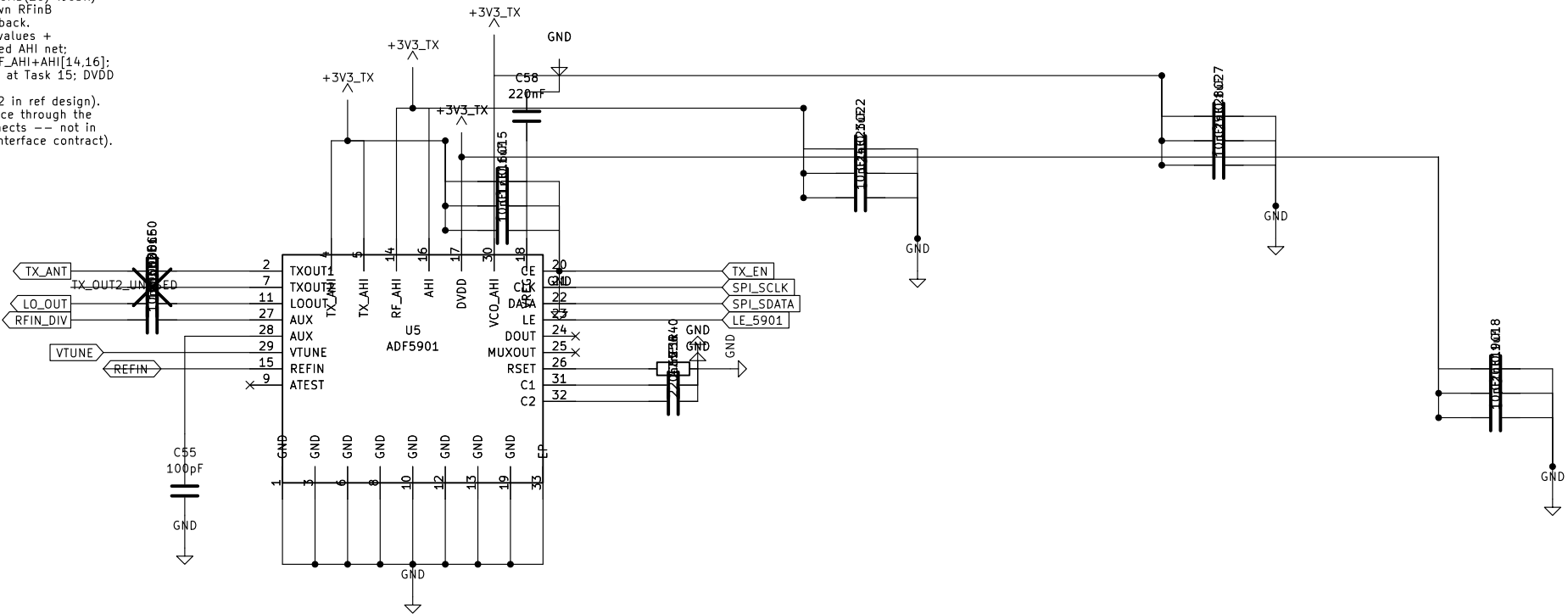


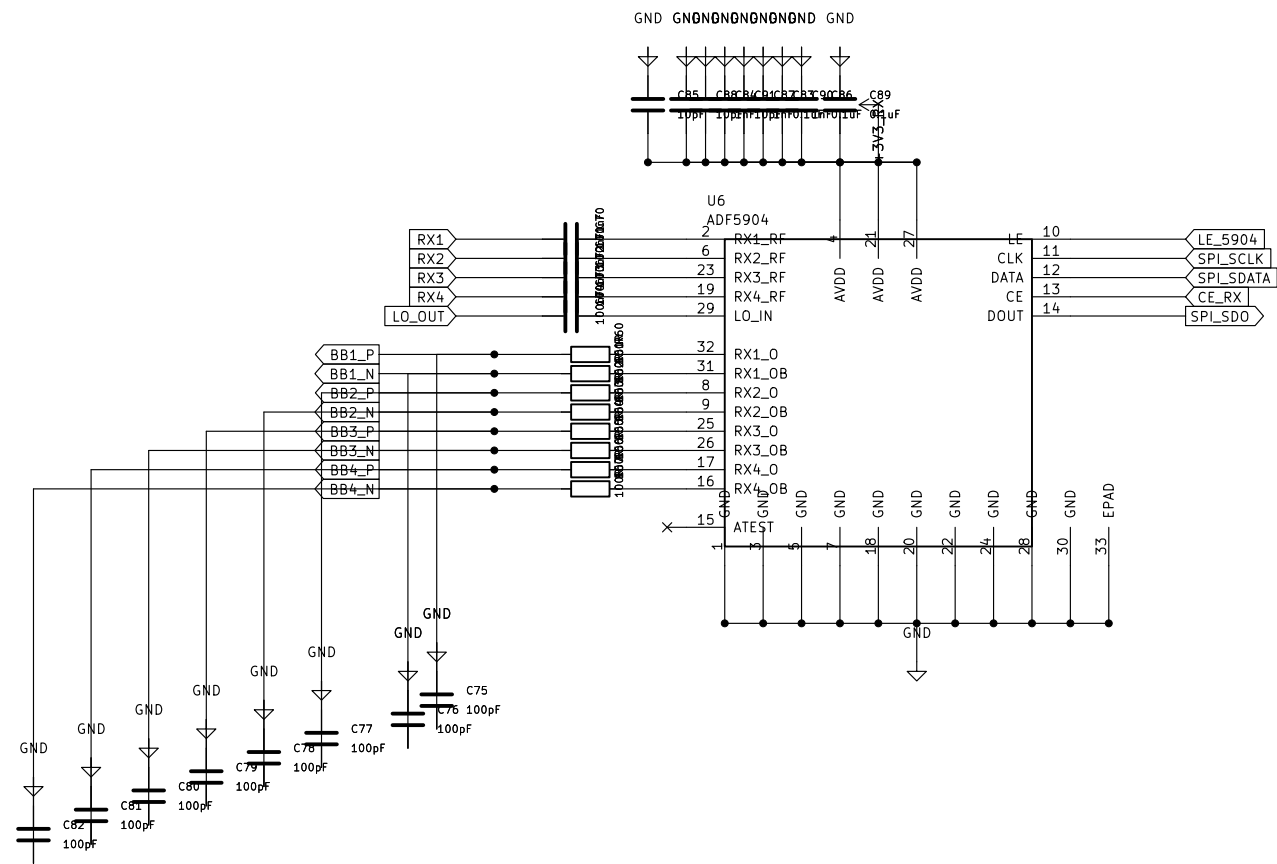
ADF4159 PLL / FMCW RAMP GENERATOR (Task 12)
AVDD/VP -> +3V3_RX (per task-11 amendment, folds ADF4159 analog onto Rx rail).
DVDD/SDVDD -> +1V8_DIG. REFIN -> Y1 100 MHz TCXO (CWX113-100.0M) via R20 1k + C30 1nF AC couple.
CP -> UG-866 Fig.12 passive 3rd-order topology, retuned for Rev C: R21=510, C33=220pF, C34=3.3nF.
R22=620, C35=47pF, R23=0 -> VTUNE. See production-package/loop-filter-analysis.json
RFinA -> RFIN_DIV (AC-coupled via C37, single-ended per datasheet note); RFinB decoupled 100nF (C36) to GND.
SPL_SDO is NOT driven from this sheet -- ADF4159 has no DOUT pin; SPL_SDO comes from ADF5904 (Task 14).



LEVEL SHIFTING (RESOLVED): ADF4159 CE/CLK/DATA/LE/MUXOUT are 1.8V-domain CMOS (abs max +2.4V, see ADF4159.md Table 4). U7 (SN74AVC4T245PW, TSSOP-16) shifts 3.3V->1.8V: SPL_SCLK/SPL_SDATA/LE_4159/CE_RX (A->port, VCCA=+3V3_RX) -> ADF4159 CLK/DATA/LE/CE (B->port, VCCB=+1V8_DIG); 1DIR/2DIR=+3V3_RX (A->B), 1OE/2OE=GND (enabled). U8 (SN74AVC2T245RSWR, UQFN-10) shifts 1.8V->3.3V: ADF4159 MUXOUT (B1) -> RAMP_SYNC (A1); 1DIR1/DIR2=GND (B->A), OE=GND. Unused U8 ch2: B2 tied GND, A2 no-connect.
100nF decoupling per supply port (C47-C50). Ports: A=VCCA=3.3V, B=VCCB=1.8V (TI SCES5761/SCES692E).
SDO: ADF4159 MUXOUT owns RAMP_SYNC only; SPL_SDO readback is ADF5904 DOUT (rx sheet) per INTERFACE.md.

ADF5901 TX SHEET (Task 13) -- integrated into the Rev C root schematic
U5 ADF5901WCCPZ. TXOUT1(2)->C50 100pF DC-block->TX_ANT (to TX antenna feed).
TXOUT2(7) UNUSED per reference design: EV-RADAR-MMIC2 Fig.10 (top block schematic) and
Fig.13 (ADF5901 pin-level schematic) route TX_OUT1 and TX_OUT2 IDENTICALLY -- each to its
own SMA jack (J8/J9), with NO dummy load or termination resistor on either output. This
sheet replicates that exactly: TX2 gets the same DC-block treatment into an open/unused
local net, not a 50-ohm termination.
LOOUT(11)->C52 100pF->LO_OUT (to ADF5904 LO_IN). AUX(27)->C53 100pF->RFIN_DIV (to ADF4159
RFinA, which is normally ac-coupled from ADF5901 AUX per adf4159.md). AUXB(28) locally
decoupled to GND (100pF) -- not routed off-sheet, mirroring ADF4159's own RFinB
-100pF to GND treatment since only single-ended AUX drives the PLL feedback.
Decoupling triads (0.1uF+1nF+10pF) per rail from adf5901.md App-circuit values +
ev-radar-mmimc2.md Fig.13 (TX_AHI: RF_AHI+AH1 combined per Fig.13's shared AHI net;
VCO_AHI separate; DVDD separate). 4 distinct +3V3_TX stubs (TX_AHI[4,5]; RF_AHI+AH1[14,16];
VCO_AHI[30]; DVDD[17]) -- merged into one +3V3_TX net by the root sheet at Task 15; DVDD
triad C18-C20 added post-review. C1-pin=47nF, C2-pin=220nF.
VREG=220nF per datasheet Table 4 pin descriptions. RSET=5.1k to GND (R22 in ref design).
REFIN(15) connects to the PLL sheet's post-coupling 100 MHz TCXO reference through the
project-global REFIN net. ATEST(9), DOUT(24), MUXOUT(25) explicit no-connects -- not in
this sheet's interface (SPI_SDO/RAMP_SYNC are sourced elsewhere per the interface contract).





Standalone sub-sheet, wired into root schematic by Task 15 Hierarchical labels: LO_OUT, RX1-4, BB1-4_P/N, SPL_SCLK/SDATA/SDO, LE_5904, CE_RX RX sheet: ADF5904 4-channel Rx downconverter + baseband coupling (Task 14) OpenFlight		
Sheet: /ADF5904 Receiver/ File: rx.kicad_sch		
Title: OpenFlight 24 GHz FMCW RF Board – Rev C – RX Sheet		
Size: A3	Date:	Rev: C
KiCad E.D.A. 10.0.4		Id: 4/4